

MILESTONE 4: CHARACTERIZING AUTOMATED REFACTORING



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YOUR PROJECT - REFACTORING – CLASS SELECTION

8. Class Selection

1. Rank all classes of the last release of the project based on Nsmells
2. Filter out classes that are too small (e.g., few and simple methods).
3. Select two classes based on your name
 1. Take the first letter of your first name (e.g., D for Davide)
 2. Take the number of the letter (e.g., D = 4)
 3. $X = \text{number} \bmod 5$ (e.g. $X = 4$)switch (X) {
 1. case 0: Classes first and last;
 2. case 1: Classes first +1 and last -1;
 3. case 2: Classes first +2 and last -2;
 4. case 3: Classes first +3 and last -3;
 5. case 4: Classes first +4 and last -4;
 6. }

YOUR PROJECT - REFACTORING – AUTOMATED REFACTORING

9. Go to Microsoft Copilot as supported by your university account. Use a prompt like the following:
 1. You are an expert Java developer. I want to improve the maintainability of the attached C_0 class ...
 2. Create C_X without changing C_0 functionality and by removing the following smells ... (report SonarCloud diagnostic)
 3. Make sure C_X passes the following tests ...
 4. Do not include in C_X changes different from what I asked. In fact, consider that C_X should replace C_0 and work with the other components of the system as C_0 currently does.
 5. This is an important request; take all the time you need to provide a complete and accurate answer to this request.

YOUR PROJECT - REFACTORING – C_X

VERSIONE DELLA CLASSE		Test Sviluppati a Partire da C_0					
		Test BB	Test CF	Test MT	Test RND	Test ES	Test LLM
C_0	ORIGINALE	N/A	N/A	N/A	N/A	N/A	N/A
C_1	Refactored	No	No	No	No	No	No
C_2	Refactored	Yes	No	No	No	No	No
C_3	Refactored	Yes	Yes	No	No	No	No
C_4	Refactored	Yes	Yes	Yes	No	No	No

RESULTS: AUTOMATED REFACTORING ANALYSIS

1. Does C_X compile? (alone or with the system)
2. Does C_X have smells? (none, old or new?)
3. Is any feature positively correlated with bugginess higher in C_X than in C_0?
 1. If yes, maybe we did not improve maintainability
4. Is any feature negatively correlated with bugginess higher in C_X than in C_0?
 1. If yes, maybe we improved maintainability